



Research paper

Vortex-induced vibration characteristic and trigger mechanism of semi-closed streamlined box girder trans-sea bridges

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ABSTRACT

In this study, sectional model wind-tunnel tests are conducted to investigate the vortex-induced vibration (VIV) of semi-closed girders with different aspect ratios. The influencing factors, including the wind attack angle and Scruton number Sc , are considered. The VIV is compared among semi-closed girder sections with three aspect ratios and unclosed widths. Numerical simulation is performed to investigate instantaneous vorticities and mean streamlines around the girder sections. The findings indicate that, at a wind attack angle of $+3^\circ$, vertical VIVs are consistently observed in the girder sections with three different aspect ratios. Moreover, a semi-closed section with a large unclosed position exhibits a vertical VIV at a wind attack angle of 0° . As Sc increases progressively, the discrepancy in the vertical VIV amplitude of the main girder diminishes gradually. At the same Sc , the vertical VIV amplitude of a girder with a smaller aspect ratio is larger than that of a girder with a larger aspect ratio. A large width of the unclosed position can deteriorate the VIV, thus affecting the flow structure surrounding the girder section. At a wind attack angle of 0° , for a girder section with a large aspect ratio and small unclosed width, the lower vortices propagate along the girder section. Additionally, the vortices propagate away from the bottom plate of the girder section when the girder section features a large unclosed width at a wind attack angle of 0° . At a wind attack angle of $+3^\circ$, the vortices at the unclosed position propagate along the section toward the leeward side. A positive wind attack angle intensifies the flow separation, and the size of the vortices at the unclosed position increases to a certain extent. These findings can provide a reference for the wind-resistant design of semi-enclosed main girders.

1. Introduction

The semi-closed box girder section, exemplified by the Pasco-Kennebec bridge, features a design that eliminates part of the bottom plate and incorporates transverse dividers aligned with the bridge's longitudinal direction, in contrast to the streamlined steel box girder (Cao et al., 2023) (Gao et al., 2018). This innovative design not only enhances overall structural efficiency but also improves constructability and durability, making it a more economical option for long-span bridges (Bai et al., 2021) (Wu et al., 2023). However, vortex-induced vibrations (VIV) pose a significant challenge and it has been observed in both wind tunnel tests and real-world environments (Zhu et al., 2023) (Li et al., 2022).

The semi-closed configuration of a box girder is relatively blunt,

which tends to promote flow separation at the semi-closed areas. When the vortex shedding frequency coincides with the natural structural frequency of the bridge, VIV of the girder sections may occur (Bai et al., 2021). VIV is a limited-amplitude vibration with a self-excited nature and it could affect the comfort of drivers and pedestrians on the bridge (Ehsan and Scanlan, 1990) (Ge et al., 2022). Although the VIV performance of such cross-sections has been studied by many scholars, VIVs phenomenon still occurs in real bridges. The Yingwuzhou Yangtze River Bridge, with main spans (850 + 850) m, consists of a concrete deck and steel longitudinal beams, experienced a vertical VIV in 2020. During VIV, the wind speed ranged from 7.0 to 7.6 m/s (Cao et al., 2023) (Zhu et al., 2021). This suggests that there are shortcomings in the current research, particularly the tendency to treat each bridge as a unique case, which results in a lack of systematic analysis. Conducting a comparative

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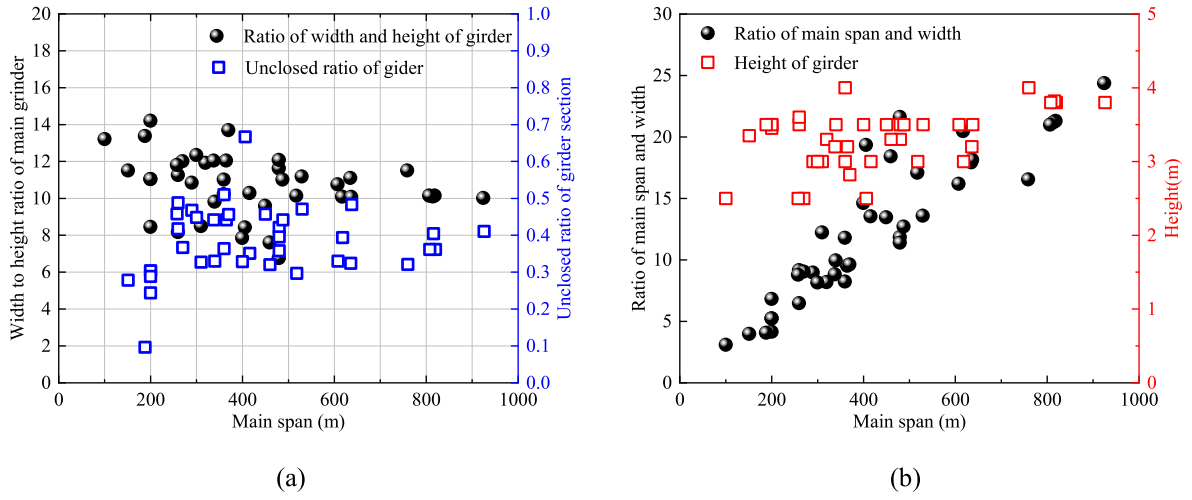


Fig. 1. Long-span bridges with semi-closed box girder sections (Meng et al., 2011; Kubo et al., 2002; Dong et al., 2021; Zhang et al., 2020; Zhu et al., 2012; Fang et al., 2017; Fujino and Yoshida, 2002).

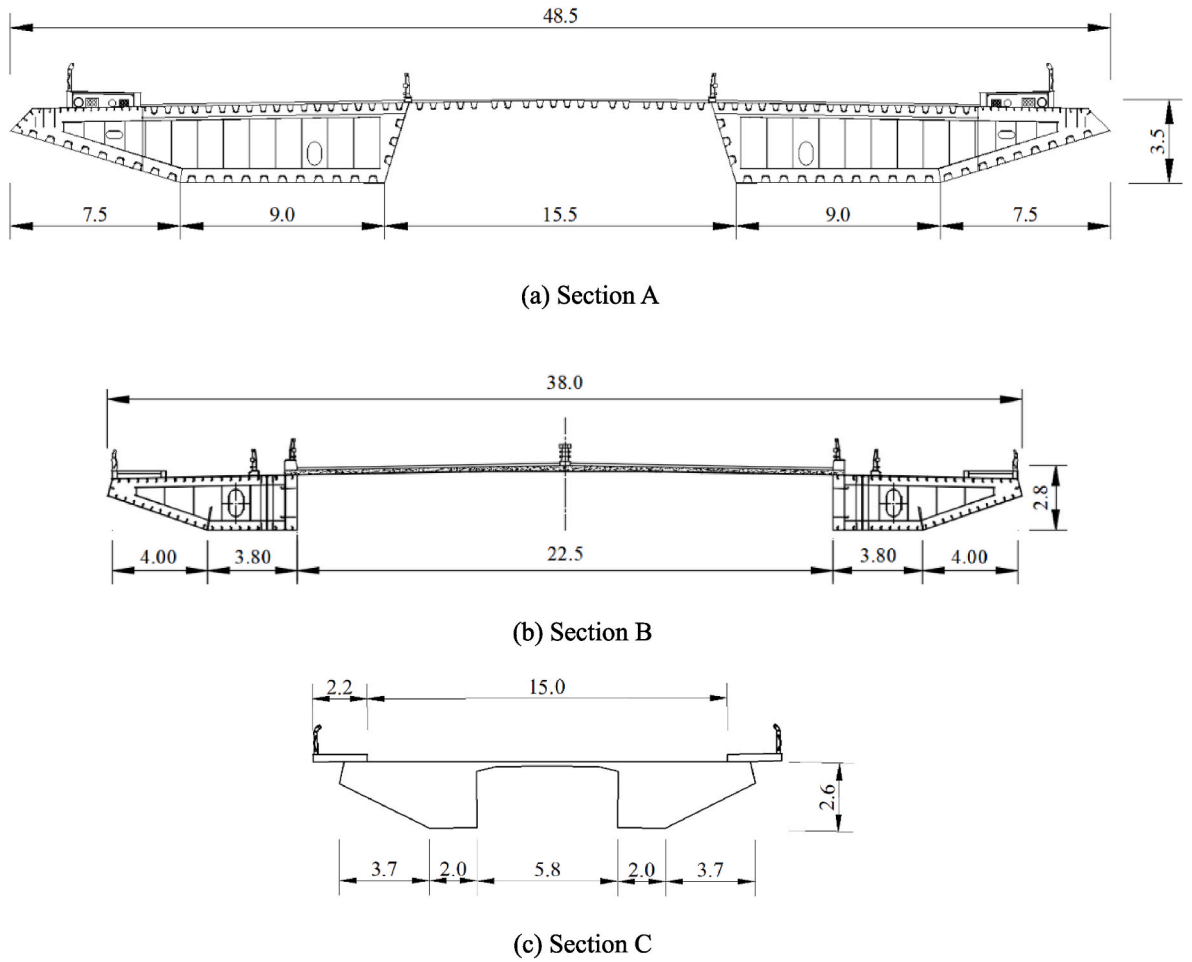


Fig. 2. Cross section of semi-closed main girders (unit: m).

study on the VIV performance of semi-closed girder sections is essential and has significant engineering implications.

Semi-closed box girder sections have been extensively used in long-span bridges (Meng et al., 2011; Kubo et al., 2002; Dong et al., 2021; Zhang et al., 2020; Zhu et al., 2012; Fang et al., 2017; Fujino and Yoshida, 2002). Fig. 1 shows the statistical findings for the long-span

bridges. As the main span of the bridge increased, the width-to-height ratio of the main girder generally exhibited a downward trend. The maximum ratio was 14, whereas the minimum ratio was approximately 7. The proportion of unclosed width in the main girder (unclosed width/main girder width) commonly falls between 0.3 and 0.5; this ratio is influenced by factors such as the torsional stiffness and other

relevant factors of girder sections. The height of the girder section is approximately 3.0–4.0 m. Generally, the width-to-height ratio and the unclosed proportion of the main girder section will influence the flow structure around the girder section and its VIV performance (Zhou et al., 2024) (Zhu et al., 2024).

The VIV suppression measures in the semi-closed box girder section encompass both mechanical and aerodynamic approaches. Traditional mechanical solutions are more suitable for box girders with greater heights (Fujino and Yoshida, 2002) (Weber and Maslanka, 2012). In recent years, innovative mechanical strategies have been proposed to address this issue (Sun et al., 2020) (Liu et al., 2024). Nonetheless, the substantial cost poses a significant challenge that needs to be considered. Therefore, adopting economical and practical aerodynamic solutions is becoming increasingly crucial. Research on the factors that induce VIV could provide insights for effective control measures against such vibrations.

Many scholars have conducted analyses on the key factors that influence VIV in semi-closed cross-sections. The bottom plate position of the opening configuration of the main girder triggers alterations in the surrounding flow structure of the semi-closed box girder section, leading to a higher likelihood of VIV in this specific cross-section (Bai et al., 2021). The wind fairing angle could improve the VIV performance of streamlined box girder sections. However, for the semi-closed box girder section, the impact of changes in the wind-fairing angle on the VIV characteristics was relatively minor (Meng et al., 2011). This is attributed to the alteration in the wind-fairing angle of the streamlined box girder, which modifies the external flow structure around the bridge section. And, this change makes it difficult for low-frequency, high-energy vortices to form when the airflow traverses a section with a small wind-fairing angle. By contrast, the external flow pattern was more intricate for the semi-closed box girder section.

Similarly, the width of the two longitudinal boxes could also influence the VIV characteristic of a π -shaped bridge girder (Kubo et al., 2002). Therefore, a semi-closed box girder section was provided without a wind fairing. Because, the aerodynamic interactions between the large-scale vortices beneath the deck and shed vortices in the wake region periodically alter the fluctuating pressure in the deck section. The VIV performance of the main girder with wind fairing can also be improved (Dong et al., 2021). The 'I' edge plate, vertical stabilizers, and horizontal flow isolating plate could also improve the VIV performance of semi-closed girders (Zhang et al., 2020). The intricate flow structure at the unclosed position continues to cause VIV issues in the main girder, which also illustrates the initiation of VIV in semi-closed box girders. Installing porous deflectors on the outer edge of the bottom plate of the main girder can suppress VIV. Porous deflectors on the windward side can effectively reduce or even eliminate the VIV in semi-closed box girders (Zhu et al., 2012). However, the VIV reduction effect of porous deflectors on the leeward side of the girder section is not as good as that on the windward side and may even exacerbate the vertical VIV amplitude (Zhu et al., 2012). Handrails on the maintenance walkway can also be a significant contributor to VIV. Therefore, countermeasures such as the installation of a horizontal wing plate and a flow-suppression plate are proposed to mitigate the amplitude of these vibrations (Fang et al., 2017).

The aforementioned studies primarily concentrate on the VIV characteristics of a specific semi-closed box girder section, leading to a 'case-by-case' methodology without comprehensive analysis. A systematic investigation into the variation law of the VIV performance in semi-closed sections cannot be established. Nevertheless, the VIV characteristics is significantly influenced by the shape of the main girder section, with the width-to-height ratio of the girder being a crucial determinant. Considering this aspect, this study investigates the effects of influencing factors such as wind attack angles and Scruton number S_c on VIV performance of three semi-closed main girder sections with varying aspect ratios. Specifically, wind tunnel tests and numerical simulation method are performed on sectional models to examine the VIV characteristics of



Fig. 3. Sectional model in a wind tunnel.

several semi-closed main girder sections, after which the variation patterns are analyzed. The flow structure around semi-closed girder sections is initially examined, laying the groundwork for engineering applications in the wind-resistance design of long-span bridges.

2. Experimental setup

2.1. Main girder section

A series of studies were conducted on three semi-closed cross-sections in engineering contexts. The three main girder cross-sections are shown in Fig. 2.

Semi-closed section A has a width of 48.50 m, height of 3.50 m, and width-to-height ratio of 13.86. The wind-fairing angle is 61° . The width of the main girder on one side is 16.50 m, the opening width is 15.50 m, and the ratio of the opening width to the main beam width is 32.0%.

Semi-closed cross-section B has a width of 38.00 m, a height of 2.80 m, and a width-to-height ratio of 13.57, which is nearly the same as that for the semi-closed section B. The wind-fairing angle is set to 95° . The width of the main girder on one side is 7.80 m, the opening width is 22.50 m, and the ratio of the opening width to the main beam width is 59.2%, which is roughly 1.85 times that in section A.

The semi-closed cross-section C has a width of 19.20 m, height of 2.60 m, and width-to-height ratio of 7.38, which is approximately one-half of that for semi-closed sections B and A. The wind-fairing angle is 105° . The width of the main girder on one side is 5.70 m, the opening width is 5.80 m, and the ratio of the opening width to the main beam width is 38.7%, which is slightly larger than that in Section A.

2.2. Sectional model and wind tunnel test

Sectional models were fabricated for wind tunnel testing to investigate the VIV characteristics of the three primary girder sections. These tests were performed at the XNJD-2 wind tunnel laboratory of Southwest Jiaotong University. The length of the sectional model of the main girder was 2.095 m. The main girder model was composed of a frame and shape. The model frame was made of high-quality wood to ensure sufficient stiffness to satisfy the requirements of a rigid model. The girder shape was refined by applying a high-performance wood skin, which simulated the size and configuration of the principal girder. The guardrails and other auxiliary facilities on the bridge deck were fabricated using plastic engineering plates and processed using an engraving machine. End plates were mounted at both ends of the sectional models to guarantee two-dimensional characteristics of the flow field surrounding the models. The endplates in the wind tunnel test were approximately $(B + D + B)$ in width and $(D + D + D)$ in height. B and D denote the width and height of the model, respectively. These sectional models were elastically suspended on an outer wall support using eight

Table 1

Main parameters of the sectional model.

Parameters	Section A		Section B		Section C	
	Real Bridge	Model	Real Bridge	Model	Real Bridge	Model
Width , B/m	48.50	0.97	38.00	0.760	17.20	0.344
Hight , D/m	3.50	0.07	2.80	0.056	2.60	0.053
Unclosed ratio of bottom plate	32.0%		59.2%		38.7%	
Equivalent mass, $m/kg \cdot m^{-1}$	37,600	15.04	41,775	16.71	31,900	12.70
Vertical frequency , f_v/Hz	0.484	1.625	0.260	2.440	0.705	5.750

springs to simulate the frequency effectively. A girder model of the wind tunnel is shown in Fig. 3.

The primary parameters of the sectional dynamic system model are listed in Table 1. The unclosed ratio of the bottom plate signifies the width of the unclosed bottom plate and the total width of the girder section. The equivalent mass of the girder model were 15.04, 16.71, and 12.70 kg/m, encompassing the mass of the girder model, supports, and springs. The required equipment mass for the models has been ensured using mass blocks. Given the similarity ratio between the actual bridge and model, the dynamic characteristics of the sectional model strictly adhere to the specifications. The wind speed ratio between the real bridge and the models can be determined by calculating the frequency

and scale factors ratios.

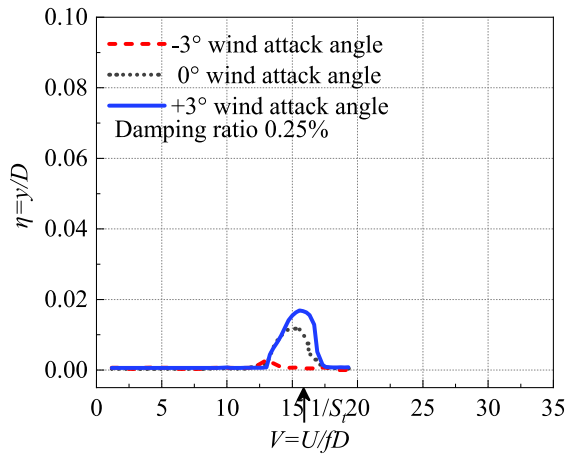
Another important parameter of the sectional model dynamic system is damping. The damping ratio includes two components: the mechanical damping ratio and the air-induced damping ratio. The damping ratios increased nonlinearly with the vibration amplitudes of the girder sections. Damping in still air contains a significant contribution from air-induced aerodynamic damping. The structural damping of a dynamic system can be increased by incorporating an oil damper or a rubber string. In addition, the maximum amplitude of the free vibration was consistent with that of the girder section during the VIV, and the damping ratio of the dynamic system was obtained.

3. VIV characteristics of semi-closed box girder

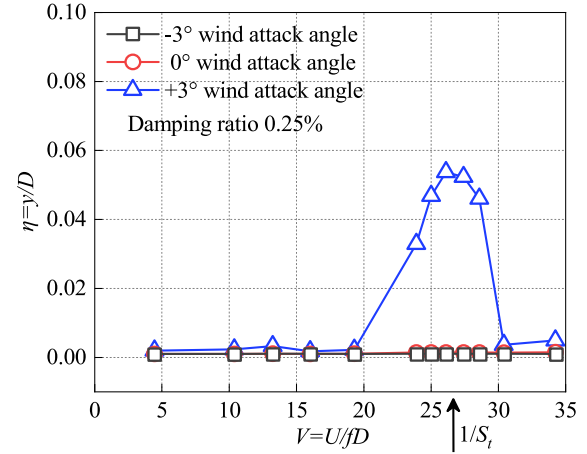
3.1. Influence of wind attack angle

Fig. 4 shows the VIV performance of girder sections under service state at wind attack angles 0° and $\pm 3^\circ$. In these figures, the horizontal axis represents the dimensionless wind speed $V = U/fD$, where U is the incoming wind speed, f is the vertical frequency, and D is the height of the girder section. The vertical axis represents the dimensionless amplitude $\eta = y/D$, where y is the vertical VIV amplitude of the real bridge.

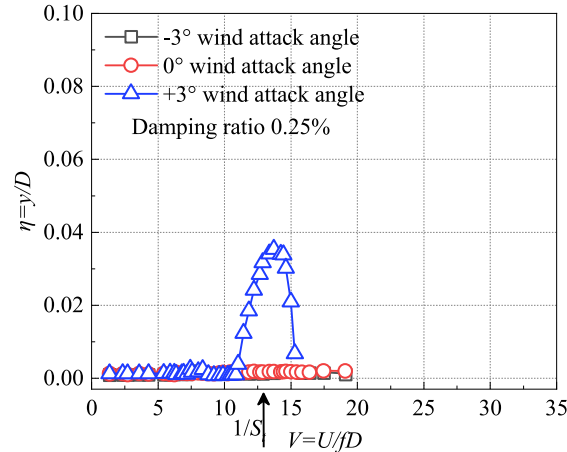
From Fig. 4(a), it could be seen that main girder A exhibits vertical VIV at wind attack angles of $+3^\circ$ and 0° , with dimensionless wind speed region of 13–17, corresponding to wind speed of 22.0–28.8 m/s in



(a) Section A



(b) Section B



(c) Section C

Fig. 4. VIV amplitude of girder sections under service status.

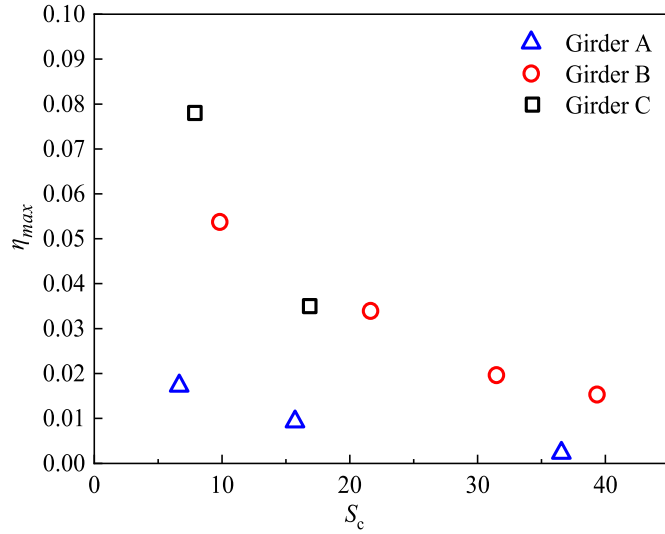


Fig. 5. Comparison of the Vertical VIV amplitudes at different S_c at wind attack angle $+3^\circ$.

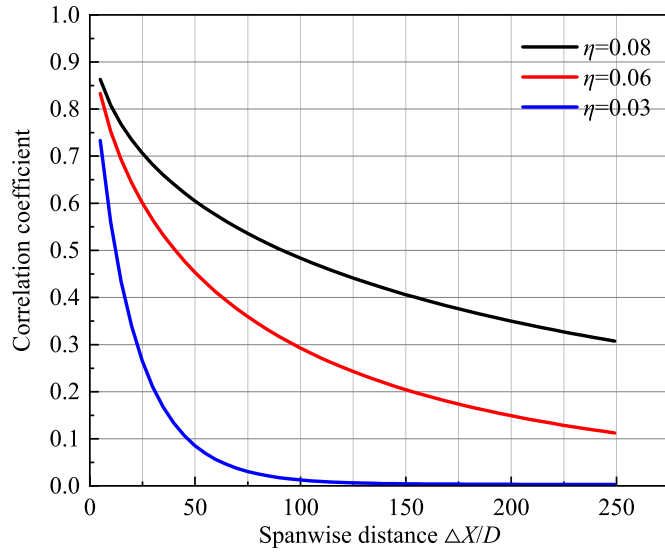


Fig. 6. Wilkinson's correlation function (Li et al., 2012).

actual. The maximum dimensionless vertical VIV amplitude is respectively 0.017 and 0.012 at $+3^\circ$ and 0° wind attack angle, with maximum amplitudes of 58.3 mm and 41.7 mm for the real bridge. No VIV occurs at a wind attack angle of -3° .

Fig. 4(b) shows that main girder B exhibits vertical VIV only at a wind attack angle of $+3^\circ$, with a dimensionless wind speed region of 20–30 corresponding to 14.6–21.0 m/s wind speeds. The maximum dimensionless vertical VIV amplitude was 0.053, with a maximum amplitude of 148.4 mm. No VIV occurred at attack angles of 0° and -3° .

Fig. 4(c) demonstrates that main girder C exhibits vertical VIV at a wind attack angle of $+3^\circ$. The non-dimensional wind speed region was 10–16, corresponding to 18.3–29.3 m/s wind speeds. The maximum non-dimensional vertical VIV amplitude for the main girder was 0.035, corresponding to a maximum amplitude of 91.5 mm. No vertical VIV is observed at wind attack angles 0° and -3° .

Although the damping ratios of the structures are similar, there are differences in parameters, such as structural quality, which necessitates further comparative analysis of the VIV performance of semi-closed box girder sections.

3.2. Influence of scruton number S_c

To further understand the influence of factors such as damping and wind attack angle on the VIV performance of semi-closed girder sections, Fig. 5 presents the variation in the maximum dimensionless vertical VIV amplitude η_{max} as the Scruton number $S_c = 4\pi M\zeta / \rho BD$ changes across wind attack angles. In the equation, M is the equivalent mass per unit length of the main girder in Table 1, ζ is the damping ratio; B and D represent the width and height of the main girder; ρ is the air density. Except for those specified above, the results in the figure correspond to $+3^\circ$ a wind attack angle.

As shown in Fig. 5, the maximum dimensionless vertical VIV amplitude of girder B decreases with increasing S_c . When the wind attack angle is $+3^\circ$ and the damping ratios are 0.25%, 0.55%, 0.80%, and 1.00%, the maximum amplitudes of the dimensionless vertical VIV are 0.054, 0.034, 0.019, and 0.015, respectively. Compared with the amplitudes for a damping ratio of 0.25%, the maximum dimensionless vertical VIV amplitudes decrease by 37%, 64%, and 72%, respectively. The maximum amplitude of the vertical VIV decreases linearly as the damping ratio increases. Under the same S_c , the vertical VIV amplitude of girder C with a smaller aspect ratio is larger, and the amplitude of girder A with a larger aspect ratio is relatively smaller. The change curves in Sections B and C overlap. As S_c gradually increases, the difference in the dimensionless vertical VIV amplitude of the main girder gradually decreases.

3.3. VIV amplitude of real bridge

The wind tunnel test results cannot be regarded as the actual VIV results for the bridge because the sectional model test did not consider the three-dimensional effects of bridges. The European Code (British Standards Institution, 2005) provides a suggested formula for the vertical VIV amplitude under crosswinds.

$$\frac{y_{max}}{D} = \frac{1}{4\pi} \frac{1}{St^2} \frac{1}{S_c} K \cdot K_w \cdot C_{lat} \quad (1)$$

where S_t is the Strouhal number $S_t = fD/U$, D the height of the main girder, f the frequency, U the incoming wind speed. The S_t values of girders A, B, and C are 0.06, 0.04, and 0.15, respectively. S_c is the Scruton number. C_{lat} represents the lateral force coefficient under a cross-wind action. K represents the modal correction factor. Assuming the modal shape is sinusoidal for suspension bridges, the modal correction factors K obtained from linear and nonlinear models are 0.10 and 0.09, respectively (Zhang and Chen, 2011).

K_w is the effective correlation length factor, which is mainly related to the cross-section length, VIV amplitude, and mode (see Fig. 6). Generally, the larger the length-to-height ratio of the structure, the smaller the reduction coefficient K_w . Taking the VIV force correlation characteristics of a square column as an example, assuming that the main span is 500 m and the main girder height is 4 m, the length-to-height ratio is 125. The reduction coefficient of the VIV amplitude is 0.25, when the maximum VIV amplitude is 0.24 m. Meanwhile, as the length-to-height ratio parameter remains unchanged and the maximum VIV amplitude decreases to 0.04 m, the reduction coefficient is nearly 0. The difference in the length-to-height ratio and transverse correlation characteristics between the sectional model and full bridge leads to a difference in the VIV amplitude. Li (Li et al., 2012) provided an iterative calculation method for the VIV amplitude of the main girder of a full bridge, which is also applicable to the calculation of results from models with different scale ratios.

According to Equation (1), the vertical VIV dimensionless amplitudes of the three main girder sections (A, B, and C) were 0.075, 0.151, and 0.024, respectively, at a wind attack angle of $+3^\circ$. However, owing to the influence of parameter values, there may have been some deviations in the results. Nevertheless, this formula can serve as a reference for

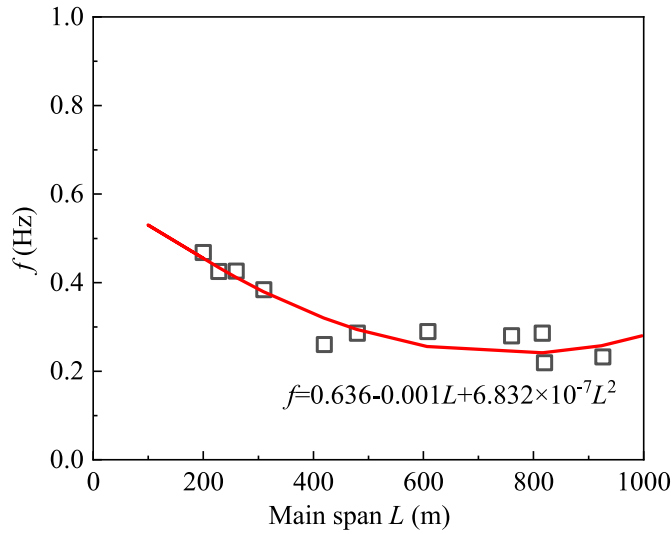


Fig. 7. First-order vertical frequencies vary with the main span.

predicting the VIV.

3.4. VIV performance of some similar semi-closed sections

Fig. 7 shows the variation in the first-order vertical frequency of the main girder with respect to the main span of the bridge structure. The horizontal axes of the figures represent the main span, and the vertical axes represent the frequency values. As shown, the frequencies of the main girder decrease nonlinearly, as the main span of the bridge structure increased. Accordingly, a nonlinear function was fitted to estimate the first-order symmetric vertical bending frequency of a semi-closed-box girder in a cable-stayed bridge, where L represents the main span of the bridge, and f represents the vertical frequency. The first-order vertical frequency of a bridge can be estimated using this formula:

To further examine the VIV performance of different semi-closed box girder sections, Table 2 summarizes the findings of several semi-closed box girders in the wind tunnel tests. The basic parameters of the cross-section and maximum VIV amplitude of the main girder cross-section are listed. As listed in Table 2, the width-to-height ratios of the main girders for several bridges fluctuated between 10° and 12° , whereas the wind fairing angles were between 60° and 70° . The inclination angles of the webs varied between 15° and 24° , and the wind speed region for the VIV ranged from 15 to 20 m/s. The maximum amplitude range for the vertical VIV of the main girder was approximately 0.3–0.5 m. Notably, some main girders underwent torsional VIV, with corresponding wind speeds exceeding 30 m/s. Therefore, these cases are not discussed herein.

Fig. 8 shows the variation in the maximum vertical VIV amplitude of the main girder with the change in Scruton number S_c at different wind attack angles. The results in the figure correspond to $+3^\circ$ and $+5^\circ$ wind

attack angle. Except for a few narrow-span main girders with smaller VIV amplitudes, the maximum VIV amplitudes of most semi-closed box girders gradually decreased as S_c increased. Meanwhile, the maximum VIV amplitude of the main girder at a $+3^\circ$ wind attack angle was slightly smaller than that at a $+5^\circ$ wind attack angle. However, it conformed to the overall change trend. This summary also revealed certain patterns that can serve as references.

4. Flow structure around girder sections

The VIV performance of the main girder without railings or other ancillary facilities disappeared. Therefore, the vertical VIV of the main girder was caused by the railing and its associated facilities, as confirmed in a previous study. Fang (Fang et al., 2017) found that when the railings were removed or when the railings on the upstream side of the incoming flow were removed, the main girder did not experience VIV. Maintenance walk railings on the upstream side of the incoming flow were the main factors for the VIV of the semi-closed box girder, and maintenance railings play an amplifying role. It has been proposed that horizontal wings or flow-blocking plates can be arranged at railings to suppress vertical VIV. Nago (Nagao et al., 1997) also believed that the railing influences the self-excited force of the vertical VIV of the main girder and the main reason for inducing VIVs is the significantly reduced damping force. In this section, the flow structure around the girder sections is analyzed.

4.1. Numerical simulation setup

The flow structure around the main girders was numerically

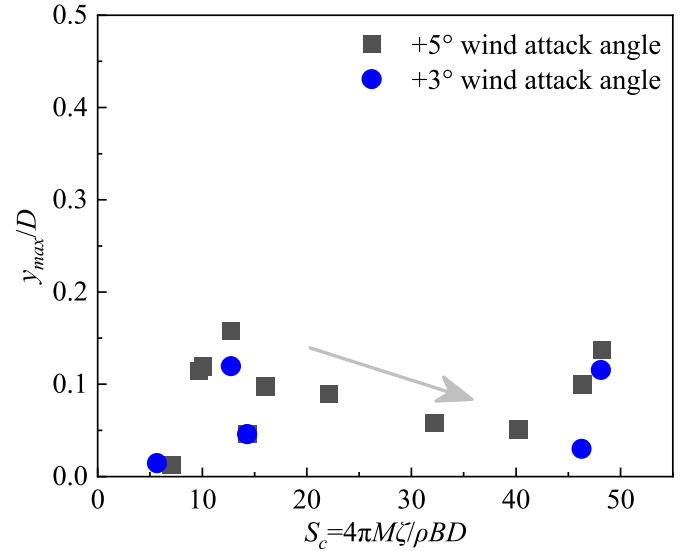
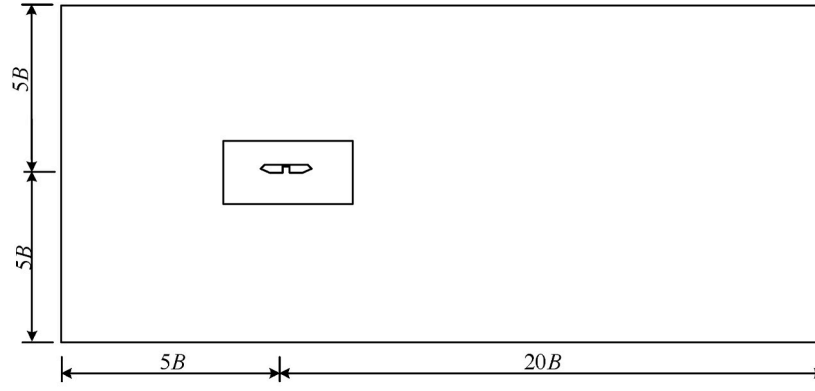


Fig. 8. Variations of maximum vertical VIV amplitudes of main girders with S_c .

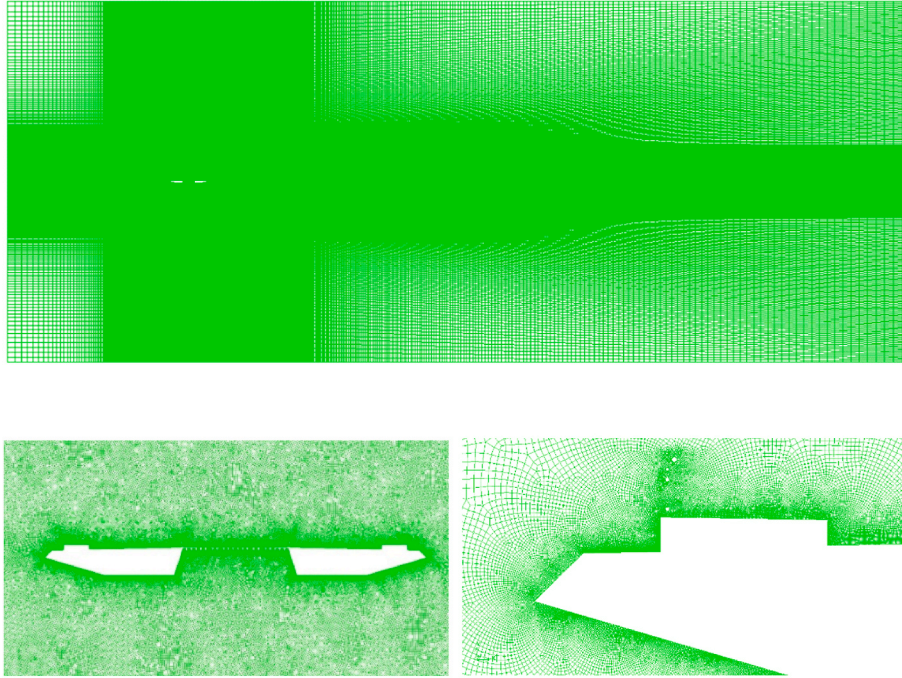
Table 2

VIV performance of some semi-closed girders ($+5^\circ$ and $+3^\circ$ wind attack angle) (Meng et al., 2011; Kubo et al., 2002; Dong et al., 2021; Zhang et al., 2020; Zhu et al., 2012; Fang et al., 2017; Fujino and Yoshida, 2002).

Number	Main span (m)	Width (m)	Height (m)	Wind fairing angle ($^\circ$)	Web inclination angle ($^\circ$)	S_c	VIV wind speed region (m/s)	Maximum VIV amplitude (m)
1	936	38.00	3.80	66	19	9.7	15.0–21.0	0.435
2	820	38.50	3.80	60	20	12.8	12.0–18.0	0.600
3	816	38.50	3.82	70	24	18.0	7–10 ($+3^\circ$)	0.270
4	608	38.6	3.5	68	18	7.5	10–12	0.790
5	480	40.63	3.50	70	17	48.2	18.0–24.0	0.480
6	480	42.26	3.50	58	14	46.4	18.0–26.0	0.350
7	310	24.17	3.00	79	20	7.0	8.3–10.8	0.036
8	200	38.60	3.50	61	15	14.3	25.2–37.5	0.160



(a) Calculation domain.



(b) Mesh

Fig. 9. Calculation domain and mesh.

simulated using FLUENT software. The scale ratio of the calculation model was the same as that of the sectional model in the wind tunnel, and the simulation was performed at wind attack angles of $+3^\circ$ and 0° . The computational domain of the numerical simulation was $25B \times 10B$ (B is the width of the main girder cross-section of the real bridge), and the vertical center of the cross-section was $5B$ and $20B$ from the front and rear boundaries, respectively. The distance between the upper and lower boundaries was $5B$. When dividing the mesh, a sparse grid was adopted in the region away from the cross-section, and mesh encryption was performed around the structural cross-section. The first layer of the grid near the cross-section had a thickness of 0.0002 m (Fig. 9), and the total number of meshes was $1,200,000$ to ensure that the y^+ value was less than 5.

Two meshing strategies were adopted to ensure grid independence. The first was the basic condition, and the second was the refined condition. For the refined condition, the number of nodes on the girder surface was twice that for the basic condition. Table 3 lists the numerical results of Strouhal number S_t of the two meshing strategies. Therefore, a basic mesh is used in this study. In addition, the time-step independence

Table 3
Numerical results of girder C.

Condition	Grid number	Time step (s)	S_t
Basic	1.2×10^6	2×10^{-3}	0.1505
Refine I	2.5×10^6	2×10^{-3}	0.1480
Refine II	1.2×10^6	1×10^{-3}	0.1503

was verified. As shown in Table 3, the results of the two-time steps showed no distinct difference. Therefore, the time step 2×10^{-3} s is adopted to balance the computing cost and result accuracy. The y^+ value and Courant number around girder section A, taken as an example, are shown in Fig. 10.

The surface boundary of the cross-section and accessory facilities was a non-slip wall boundary, and the inlet had a uniform flow velocity. The outlet was a pressure outlet boundary condition, and the upper and lower boundaries were wall boundary conditions. An *SST k- ω DES* (Delayed detached-eddy) turbulence model was adopted. The equation set

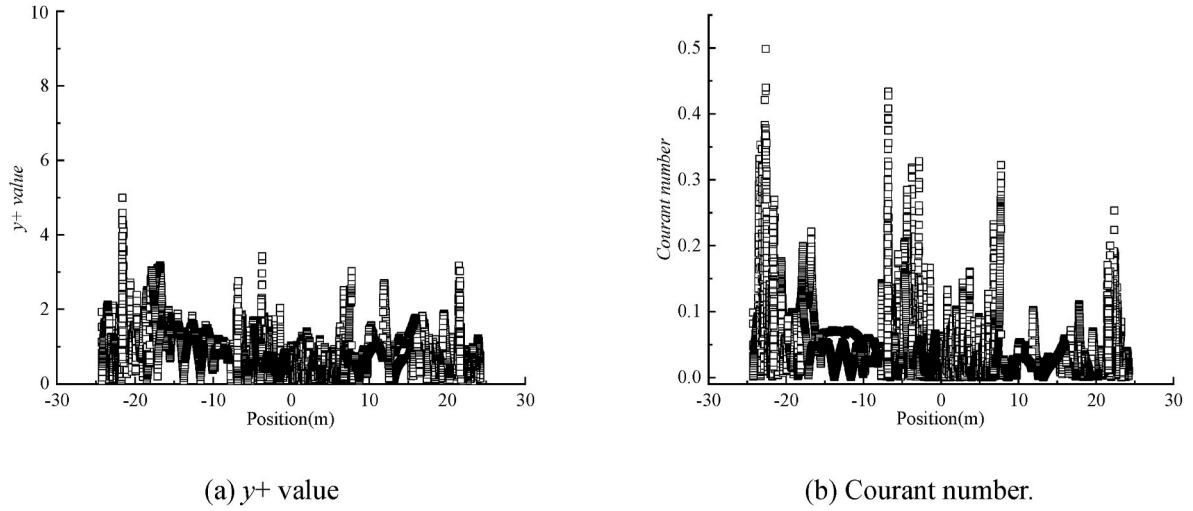
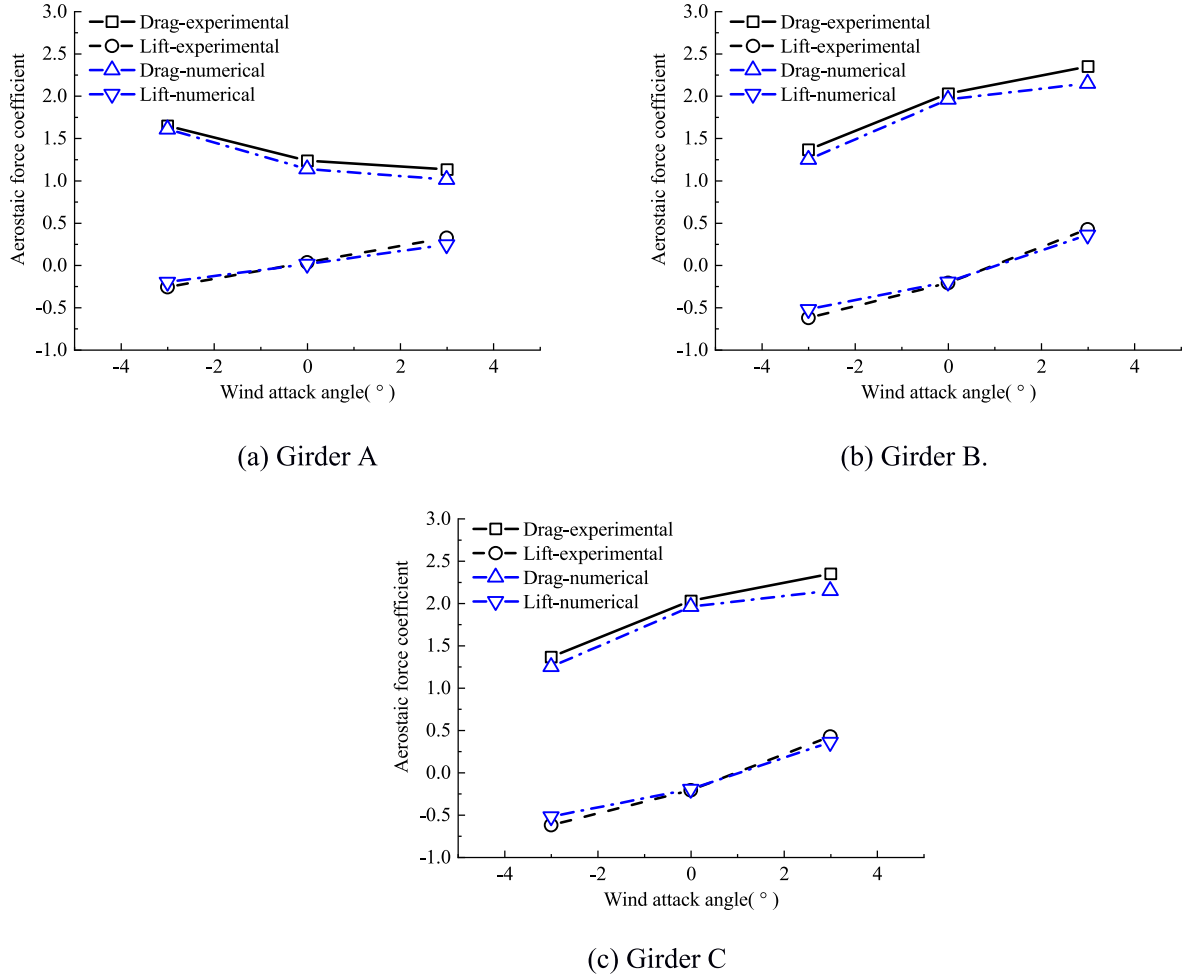
Fig. 10. Y^+ values and courant number.

Fig. 11. Comparison of numerical and experimental aerodynamic coefficients.

was solved using the SIMPLE algorithm, and the convergence term residual was controlled at 1×10^{-6} . The time step was set to 2.0×10^{-3} s to ensure the Courant number was less than 1 (Fig. 10(b)). The Reynolds number was consistent with that of the main girder sectional model.

To obtain more comparative results, the numerical study primarily investigated the flow field around the bridge deck at the wind attack

angles of 0° and $+3^\circ$. The simulation was validated by comparing the numerical and experimental values of the aerodynamic coefficients associated with the bridge deck (see Fig. 11). These findings demonstrate that the mean aerostatic-force coefficients derived from the numerical simulation closely align with the experimental results. The mean drag force and lift force coefficients could be represented by C_D and C_L ,

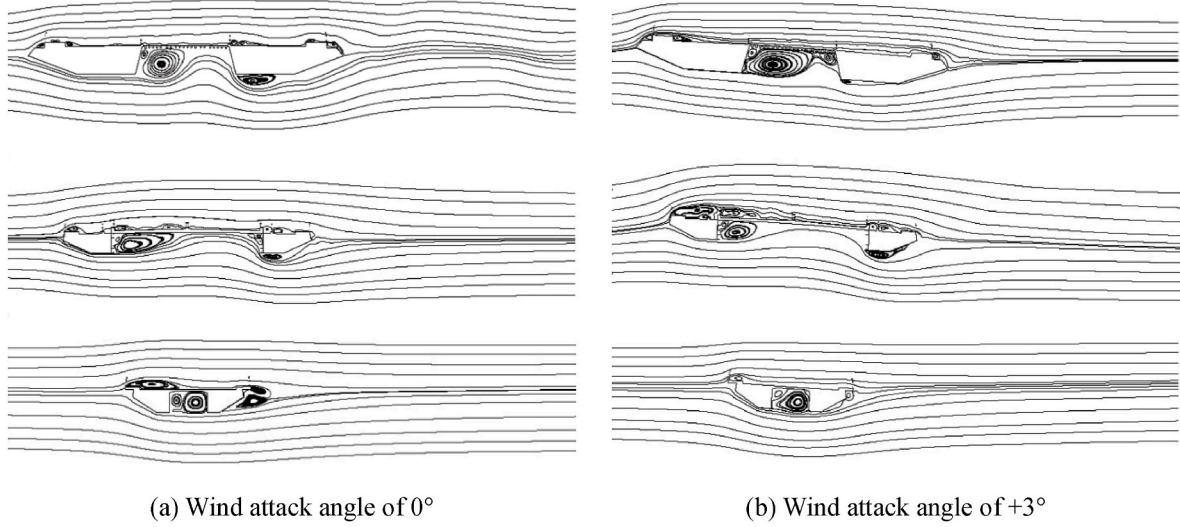


Fig. 12. Mean streamlines around girder sections.

as is shown in Equation (2).

$$\begin{aligned} C_D &= F_D / (0.5\rho\bar{U}^2 D) \\ C_L &= F_L / (0.5\rho\bar{U}^2 B) \end{aligned} \quad (2)$$

Where F_D and F_L represent the mean drag force and lift force on girder section. ρ is the air density and $\bar{U}$ represents the incoming mean wind speed. The width and height of the girder section is represented by the B and D .

4.2. Mean streamlines around girder sections

Fig. 12 shows the mean streamlines around the girder sections at the wind attack angles of 0° and $+3^\circ$. Generally, smaller vortices are present on the windward sides of bridge decks. The main distinction lies in the semi-closed position, windward side of the bridge deck, and lower deck of the girder section.

In Fig. 12, large vortices emerge at the unclosed position for girder section A. By comparing the flow structures at wind attack angles of 0° and $+3^\circ$, it is evident that a positive wind attack angle can lead to the

formation of larger vortices at the unclosed positions and relatively smaller vortices at the bottom plate on the leeward side of the girder section. Separation bubbles thicken the boundary layer, increasing the drag force on the cross-section. The increase in the size of the vortices at the unclosed position leads to a pressure difference between the upper and lower surfaces of the cross-section, increasing the lift force.

For girder section B, the vortices at the semi-closed position remain approximately unchanged at the wind attack angles of 0° and $+3^\circ$. There are relatively large vortices on the windward side of the bridge deck and a bottom plate on the leeward side of the bridge girder at a wind attack angle of $+3^\circ$. This is the main reason for the increase in the drag force in Section B at a wind attack angle of $+3^\circ$. The vortex size increases on the bottom plate of the leeward side, directly leading to a pressure difference between the upper and lower sections, and the average lift increases accordingly.

For girder C, large vortices remained at the semi-closed positions. At a wind attack angle of 0° , large vortices were observed at the windward side of the bridge deck and the tail region of the bridge girder. However, only large vortices were observed at the unclosed position under a wind attack angle of $+3^\circ$. The vortex size at the unclosed section increased,

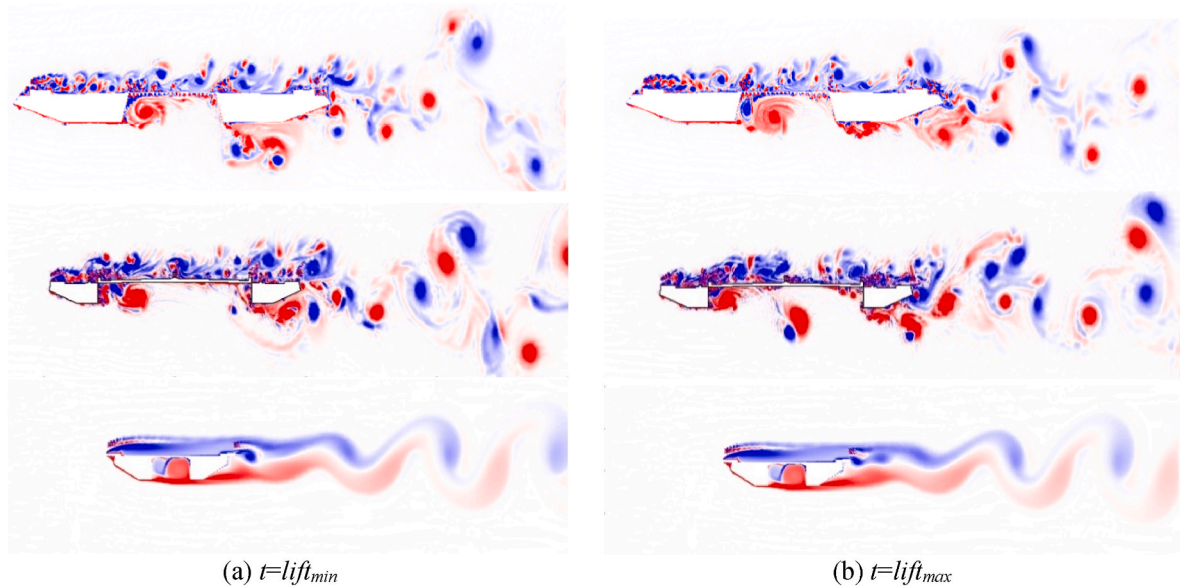


Fig. 13. Instantaneous flow structure around the girder section at wind attack angle of 0° .

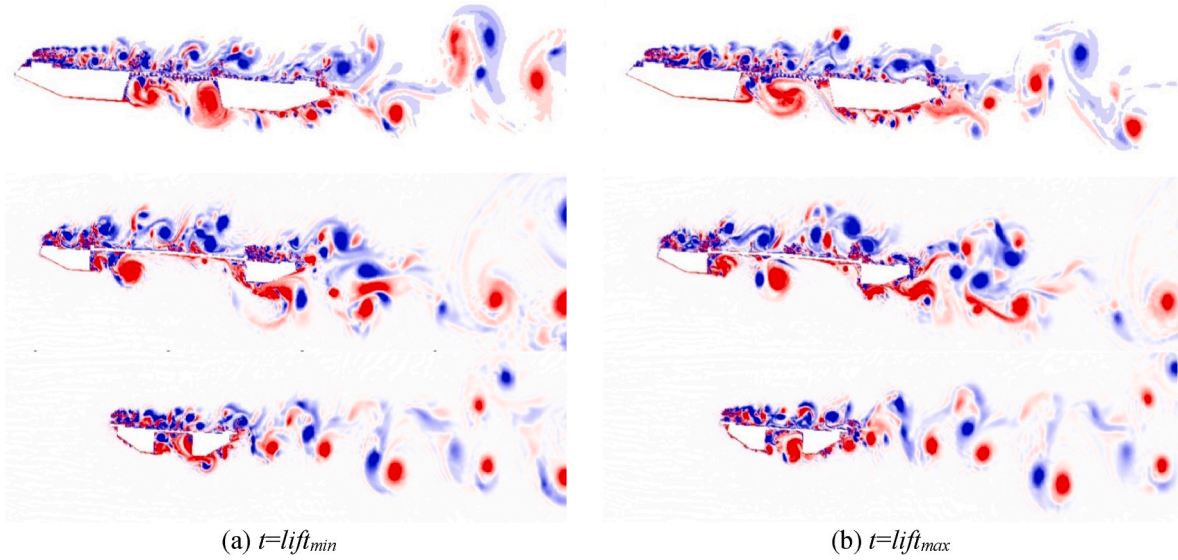


Fig. 14. Instantaneous flow structure around girder section at the wind attack angle of $+3^\circ$.

and the vortices on the bridge deck were eliminated, which resulted in an increase in the pressure difference between the upper and lower surfaces of the cross-section; consequently, the lift force increased under a wind attack angle of $+3^\circ$.

4.3. Instantaneous vorticity around girder sections

Figs. 13 and 14 show the instantaneous vorticity diagrams around the three girder sections. The diagram shows the maximum and minimum lifting moments of the girder at wind attack angles of 0° and $+3^\circ$, which are represented by $t = lift_{max}$ and $t = lift_{min}$. The blue and red colors represent the clockwise and counterclockwise vortices, respectively. The vorticity (w_z) in the z -direction can be written as

$$w_z = \frac{\partial V}{\partial x} - \frac{\partial U}{\partial y} \quad (3)$$

where x and y denote the coordinates along the inflow and cross-flow directions, respectively.

Generally, in the case of an initial girder section featuring railings, the airflow experiences separation at the windward edge and railings, resulting in the formation of upper vortices. A relatively large separation vortex was observed at the railing base. After passing through the railings, the airflow was separated again, and vortices were generated and reattached to the bridge deck. The size of the vortices on the bridge deck gradually increased. The lower airflow moved along the inclined web plate under the main girder. As the airflow continued, it passed through the semi-closed position and crashed at the bottom of the downstream section. Here, the airflow experienced backflow, resulting in the formation of large vortices. Eventually, the vortices located above and below the girder section were shifted towards the end of the girder section and then shed. This shedding process resulted in a periodic lift, which, in turn, induced VIV.

At a wind attack angle of 0° (Fig. 13) for girder A, the vortices at the semi-closed position crashed into the leeward-side girder section. Vortex shedding occurs in the wake of the main beam. However, for girder B, the vortices gradually moved away from the main beam because of the sufficient length of the semi-enclosed position. Only a small portion of the airflow collided with the leeward side of the main girder, producing vortices in the downstream main girder that moved along the lower bottom plate of the main girder. Vortex shedding occurs in the wake of the main beam, and the wake width is relatively wide. Comparing girder A and B, the smaller unclosed width at the bottom plate of the main

beam makes the section more streamlined, resulting in a narrower width of the wake. However, this also leads to the presence of larger vortices at the bottom plate, which causes VIV in the section. For girder C, most of the airflow moved along the upper and lower sides of the main girder and towards the tail wake region of the girder sections. The periodic shedding of vortices from the wake of girder is a typical 'Kármán vortex' pattern.

At a wind attack angle of $+3^\circ$ (Fig. 14), the positive wind attack angle intensifies flow separation. The size of the vortices on the bridge deck increased significantly compared to that at a 0° wind attack angle. This was more pronounced in girder section C. The size of vortices on bridge deck gradually increases and moves to the leeward side. Meanwhile, the vortex size at the unclosed position below the main girder increased significantly at wind attack angle $+3^\circ$. And, the vortices at bottom slab also gradually move to the leeward side, colliding with the box girder on the leeward side, and then move along the inclined web of the main girder. The vortices on the bridge deck and at the bottom slab experience periodic vortex shedding in the wake of the main beam. For girder A, the vortices at unsealed bottom plate of the main beam significantly increase, and the wake width behind the main beam become wider, which results in VIV occurring in the section at wind attack angle of $+3^\circ$. In girder section B, the vortices at the unclosed position did not move far from the section but moved towards the leeward side of the girder section. The size of the vortices on bridge deck is increased. This could be the reason for the VIV of the girder section B at a wind attack angle of $+3^\circ$. Large-scale vortices appear at the bridge deck and the bottom plate of the girder C. The width of the tail flow of the main girder was also widened, especially in girder section C.

The three main girder cross-sections exhibit distinct flow patterns: Pattern A, characterized by moving vortices at bridge deck; Pattern B, featuring moving vortices at bottom slab; and Pattern C, which involves periodic vortex shedding in the wake of the main beam. Girders A and B have same width-to-height ratio but differ in the unclosed width. At a wind attack angle of 0° , the flow structure around girders A and B (with small and large unsealed bottom plates, respectively) is predominantly Pattern B, while the flow structure around girder C is primarily Pattern C. At a wind attack angle of $+3^\circ$, the flow structure around all three girder sections exhibits a combination of Patterns A, B, and C. Patterns A and B are the primary contributors to VIV. The flow field structure around cross-section A with smaller unclosed width is mainly composed of Patterns B and C, whereas the flow field structure around cross-section with larger unclosed width encompasses Patterns A, B, and C. The wind fairing could influence the vortices on bridge deck.

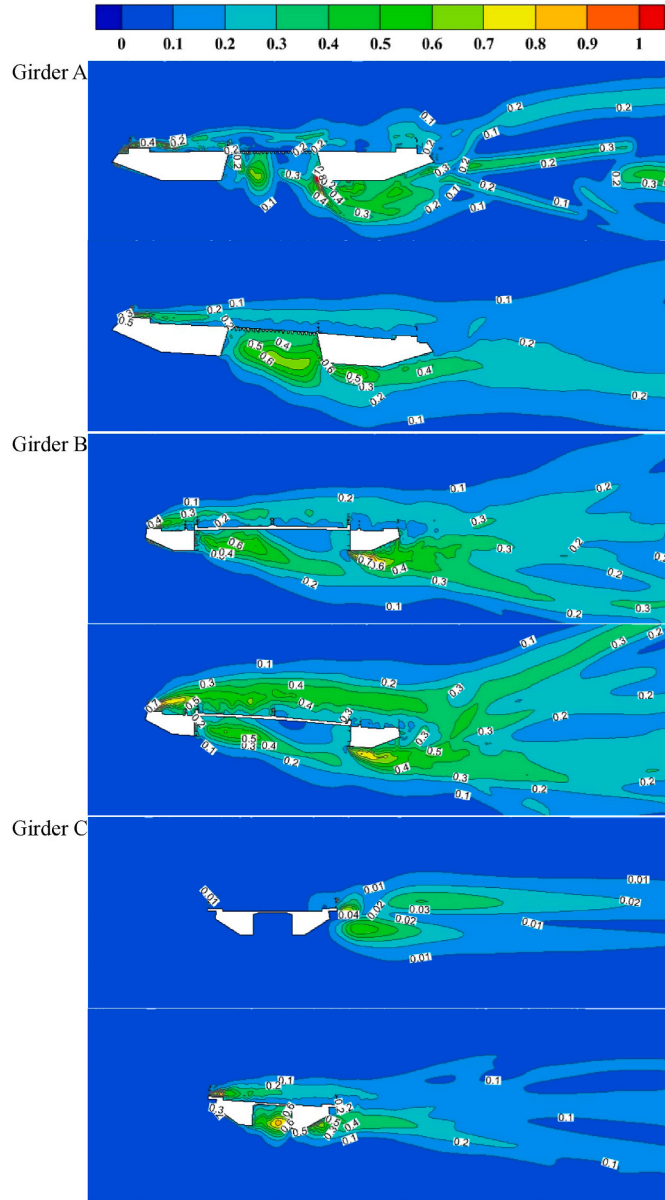


Fig. 15. RMS pressure distribution around the girder section at the wind attack angles of 0° and $+3^\circ$.

Fig. 15 shows the fluctuating pressure distribution around the three girder sections at wind attack angles of 0° and $+3^\circ$. The vortex pair shedding in the weak region at the wind attack angle of $+3^\circ$ is wider. The aerodynamic interactions occurring in the wake zone between the shedding vortices and the large-scale vortices beneath the deck periodically alter the pressure on the section surface, leading to VIV. If the vortex-shedding frequency aligns with that of the bridge deck, the vortex-shedding process becomes entrained by the deck oscillations, potentially amplifying the aerodynamic interaction and energy conversion. This aerodynamic instability is referred to as the lock-in phenomenon, with periodic oscillations characterized by VIV.

A positive wind attack angle enhances the fluctuation pressure at the unclosed position for girder A, with a larger aspect ratio and smaller unclosed width. Meanwhile, the fluctuation pressure at the bottom plate of the leeward side and tail region increases in girder A. At a wind attack angle of 0° , the vortices alternate in shedding at the unclosed position and reattached to the bottom plate of the leeward side of the main girder, resulting in a larger fluctuation pressure at the bottom plate region. The pressure fluctuation on bridge deck is relatively small. A

positive attack angle enhances the pressure fluctuation around the girder section with a smaller aspect ratio.

For girder B, larger unclosed widths result in relatively larger fluctuation pressures at the unclosed position. A positive attack angle significantly increases the fluctuation pressure on bridge deck. The tail region intensity and width of the main girder are also enhanced at a wind attack angle of $+3^\circ$. This proves that at a wind attack angle of $+3^\circ$, flow structure with Pattern A, B, and C all exist around the girder section B.

For girder C, a positive attack angle significantly increases the fluctuation pressure around the bridge girder, including the bridge deck, bottom plate and tail region of the girder section. Therefore, there is a vertical VIV at a wind attack angle of $+3^\circ$ and no vertical VIV at a wind attack angle of 0° . This proves that at a wind attack angle of $+3^\circ$, flow structure with Pattern A, B, and C all exist around the girder section C.

Fig. 16 presents the fluctuating lift coefficients for the three girder sections, while the mean force coefficients are not displayed. At a wind attack angle of 0° , the fluctuation strength of the lift coefficients for Girder A is notably higher, which aligns with the observed flow structure and vortex-induced vibration (VIV) performance of Girder A. Conversely, at a wind attack angle of $+3^\circ$, the fluctuation lift coefficients for Girder B exhibit greater susceptibility, whereas those for Girder C are comparatively weaker. Fig. 17 illustrates the Fourier transform of the time-series curve of the lift coefficient. The Strouhal numbers (St) are measured at 0.074, 0.050, and 0.091 for Girders A, B, and C, respectively. These results are consistent with findings from wind tunnel testing, further confirming the reliability of the numerical analysis.

The fluctuation lift force coefficients C_L' is the Root Mean Square (RMS) values of fluctuation lift force coefficients. Fig. 18 illustrates the RMS values of lift coefficients at wind attack angles of 0° and $+3^\circ$. The variation in fluctuation lift coefficients among different girder sections is substantial. At a wind attack angle of $+3^\circ$, the fluctuation lift force coefficients for girders B and C are lower compared to their values at a wind attack angle of 0° . At a wind attack angle of 0° , the fluctuation lift force coefficients for girders B and C are relatively smaller. In contrast, at a wind attack angle of $+3^\circ$, the fluctuation lift force coefficients for girder B are greater than those for girders A and C. Overall, the RMS fluctuation lift coefficients effectively reflect the trends in fluctuation lift and provide valuable insights into the vortex-induced vibration (VIV) performance of the girder sections.

4.4. Discussion

Owing to the influence of the parameter, some deviations may occur in the estimated results. Nevertheless, the prediction formula can still serve as a reference for predicting VIV. Except for a few narrow-span main beams with smaller VIV amplitudes, the maximum VIV amplitudes of most semi-closed box girders gradually decreased with an increase in the S_c number, showing a nearly linear change and providing a good basis for estimating the amplitude values of the VIV of bridge girders of the same type.

The aspect ratio of the girder section and the width of the unclosed position influence the flow structure around the girder section. At a wind attack angle of 0° , for a girder section with a larger aspect ratio and smaller unclosed width, the lower vortices move along the girder section. In addition, the separation effect is relatively weak. At a positive wind attack angle, the separation effect of the airflow is significantly enhanced, even for girder sections with smaller aspect ratios. The vortices move away from the bottom plate of the girder section when the girder exhibits a larger unclosed width. The vortices around the girder sections and the width of the wake also increase.

The three main girder cross-sections exhibit distinct flow patterns: Pattern A, characterized by moving vortices at bridge deck; Pattern B, featuring moving vortices at bottom slab; and Pattern C, which involves periodic vortex shedding in the wake of the main beam. Girders section with same width-to-height ratio but differ in the unclosed width. At a

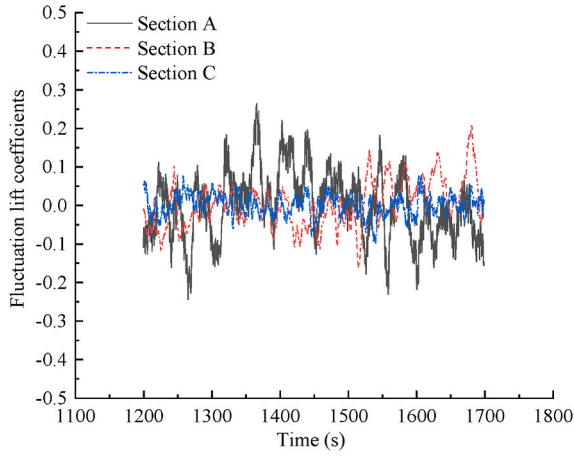
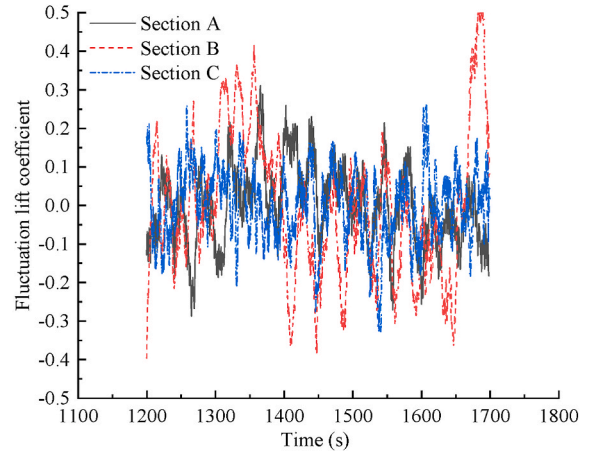
(a) Wind attack angle of 0° (b) Wind attack angle of $+3^\circ$

Fig. 16. Fluctuation lift coefficients.

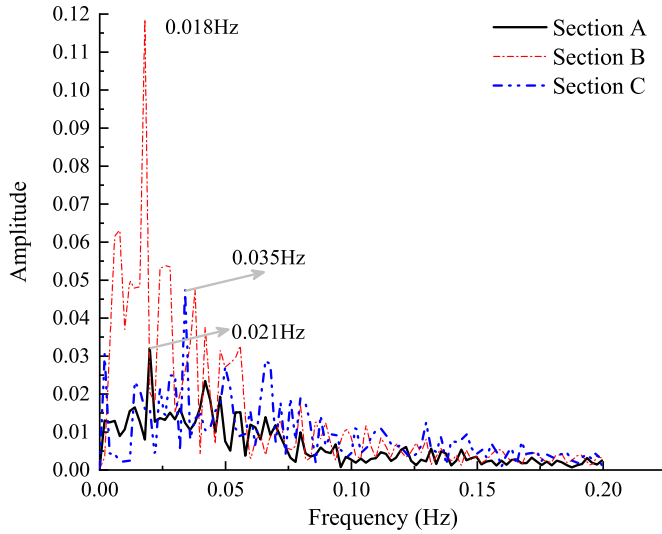


Fig. 17. Fluctuation lift coefficients.

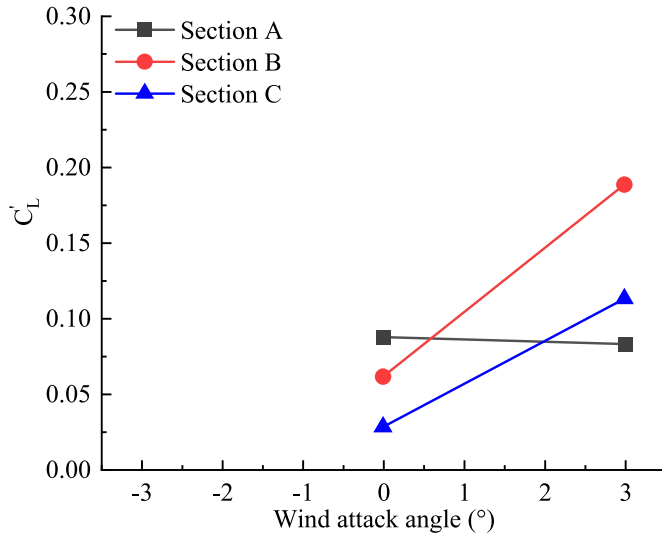


Fig. 18. RMS values of Fluctuation lift coefficients.

wind attack angle of 0° , the flow structure around girders (with small and large unsealed bottom plates, respectively) is predominantly Pattern B, while the flow structure around girder section with smaller aspect ratio is primarily Pattern C. At a wind attack angle of $+3^\circ$, the flow structure around all three girder sections exhibits a combination of Patterns A, B, and C. Patterns A and B are the primary contributors to VIV. A relatively larger unclosed width could make the flow structure around girder sections more complex.

5. Conclusion

Based on sectional model wind tunnel tests and numerical simulations, the VIV characteristics of three different semi-closed girder sections and their corresponding flow structures were investigated. In addition, semi-closed girder sections with varying aspect ratios and unclosed widths were compared. The influences of wind attack angle and Scruton number S_c were considered.

At a $+3^\circ$ wind attack angle, vertical VIV was consistently observed in the semi-closed girder sections. The vertical VIV amplitude of the main girder gradually decreased as the S_c values increased. The maximum amplitude of the vertical VIV decreased linearly as the damping ratio increased. Under the same S_c , the dimensionless vertical VIV amplitudes of the girders with an aspect ratio of 7.38 were larger, and the amplitude of the girder with an aspect ratio of 13.86 was relatively smaller. As the S_c value gradually increased, the difference in the dimensionless vertical VIV amplitude of the main girder gradually decreased. A wider unclosed width led to worse VIV performance of the main girder.

Generally, at a wind attack angle of 0° for a girder section with a larger aspect ratio and smaller unclosed width, the lower vortices moved along the girder section. When the unclosed position widened, the vortices moved away from the bottom plate of the girder. A positive wind attack angle intensified the flow separation, and the size of the vortices at the unclosed position increased to a certain extent. In addition, the vortices at the unclosed position moved along the main girder towards the leeward side. The three main girder cross-sections exhibit distinct flow patterns: Pattern A, characterized by moving vortices at bridge deck; Pattern B, featuring moving vortices at bottom slab; and Pattern C, which involves periodic vortex shedding in the wake of the main beam. The width-to-height ratio and unclosed width would influence the flow pattern around girder section. A larger unclosed width would influence the flow structure and which is more complex. At $+3^\circ$ wind attack angle, flow structure around girder section with smaller width-to-height ratio is pattern A, B and C which induce the VIV.

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CRediT authorship contribution statement

Qingsong Duan: Writing-original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Cunming Ma:** Writing-review & editing, Investigation, Project administration, Funding acquisition. **Tao Zhang:** Investigation, Formal analysis, Data curation. **Zhiguo Li:** Investigation, Formal analysis.

Declaration of competing interest

No potential conflict of interest was reported by the author(s).

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